	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence C5, C6 and C7** **Continuous distributions and sample data** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.	**Assessment criteria:** C5: Calculates and interprets the mean, variance and standard deviation of continuous distributions using standard formulas and functions. C6: Estimates interval probabilities using relative frequencies from generated samples. and compares them with theoretical probabilities. C7: Calculates sample means, variances, and standard deviations for different sample sizes and data representations.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C5, C6, and C7 separately (6 points each; 18 total). Each task specifies its points. Partial credit requires correct procedures; answers without procedures receive zero. Missing names or late submissions incur a 1-point deduction per criterion.**Passing requires at least 4 points per criterion**. Use unrounded intermediate values and round final statistics to 3 decimals and percentage errors to 2.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	1	–
6	–	2	–
7	–	1	–
8	–	–	2
9	–	–	1
10	–	–	1
11	–	–	2
Total	**6**	**6**	**6**

1. Two custom-print studios [C5-2]

Two studios print custom posters, with each order taking between 3 and 15 hours to complete. Both studios handle a wide variety of orders, but their production processes differ. Studio I specializes in large, detailed posters that require extensive customization. As a result, longer completion times are more common, with orders becoming progressively more likely to require additional production time. Studio D uses a more efficient printing process designed for quick orders. Shorter completion times are more common, and orders become progressively less likely to require additional production time. Let I and D represent the respective completion times in hours. Identify each distribution from the contextual information and calculate its mean, variance, and standard deviation using the corresponding formulas. **[2]**

2. Bookings and daily takings [C5-2]

A camera repair shop's daily revenue varies due to many small, independent factors, including customer demand and repair costs. Most days, revenue is close to \$3200, while higher and lower amounts become increasingly uncommon at similar rates. The standard deviation is \$200. The shop receives online bookings independently at an average rate of 10 per hour. Bookings arrive at a steady average rate, and the time since the last booking does not affect when the next arrives. Identify the distributions for revenue and time between bookings and calculate their means, variances, and standard deviations, expressing revenue in dollars and waiting time in minutes. Interpret the mean revenue. **[2]**

3. Two parcel-labeling stations [C5-2]

A delivery company uses two parcel-labeling stations. At station A, an automated machine labels parcels in 5 to 19 minutes. Due to differences in parcel sizes and handling requirements, completion times vary unpredictably, with no particular duration being more common than another.

At station B, staff label special parcels, also taking between 5 and 19 minutes. Most parcels are completed in around 10 minutes, while shorter and longer processing times become progressively less common. Very quick jobs are especially rare, and longer jobs occasionally require additional handling.

Identify the distributions for the labeling times in both stations and calculate their respective means, variances, and standard deviations using the corresponding formulas. **[2]**

4. File upload times, grouped data [C6-2]

A cloud storage company monitors the time required for users to upload files to its servers. To evaluate the performance of its system, the company records the upload times, in seconds, for a sample of 24 files. The results are summarized in the following table, where each class includes its lower bound but excludes its upper bound.

Class	$[2, 8)$	$[8, 14)$	$[14, 20)$	$[20, 26)$
f_i	5	9	7	3

According to the company's theoretical model, the probability that a file takes between 8 seconds and 20 seconds to upload is 0.75. The company wants to assess how closely the observed results agree with this prediction. Calculate the empirical probability of this interval and determine its absolute and percentage errors relative to the theoretical probability. **[2]**

5. Office electricity use, raw data [C6-1]

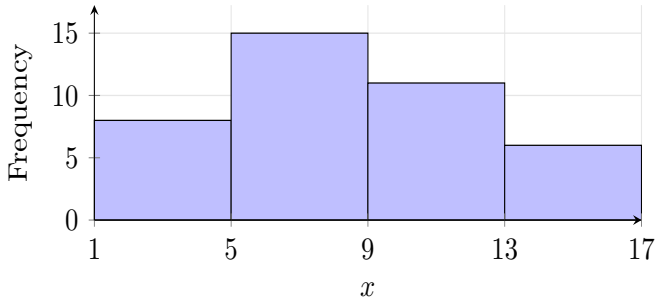
An office building monitors its daily electricity consumption to evaluate how accurately its energy forecasting model predicts actual usage. The building manager records electricity consumption over 20 working days, measured in tens of kilowatt-hours. The observations are

1.3	1.7	2.0	2.2	2.6	2.9	3.3	3.5	3.8	4.2
1.6	1.9	2.3	2.7	3.1	3.4	3.6	3.9	4.4	4.6

According to the building's theoretical energy model, the probability that daily electricity consumption falls between 20 and 35 kWh, including both limits, is 0.60. The manager wants to determine how closely the observed consumption agrees with this prediction. Calculate the empirical probability of this interval and determine its absolute and percentage errors relative to the theoretical probability. **[1]**

6. Museum entrance queues, histogram data [C6-2]

A museum monitors how long visitors wait in line before entering to evaluate the accuracy of its queue management model. During a busy weekend, staff record the waiting times of 40 visitors, measured in minutes. The original observations are no longer available, but their frequencies are preserved in the following histogram. Each bar represents the number of visitors whose waiting times fall within the corresponding interval, including the lower bound but excluding the upper bound.



According to the museum's theoretical model, the probability that a visitor waits at least 5 minutes but less than 13 minutes is 0.70. The museum wants to assess how closely the observed waiting times agree with this prediction. Calculate the empirical probability of this interval and determine its absolute and percentage errors relative to the theoretical probability. **[2]**

7. Equipment rental times, frequency data [C6-1]

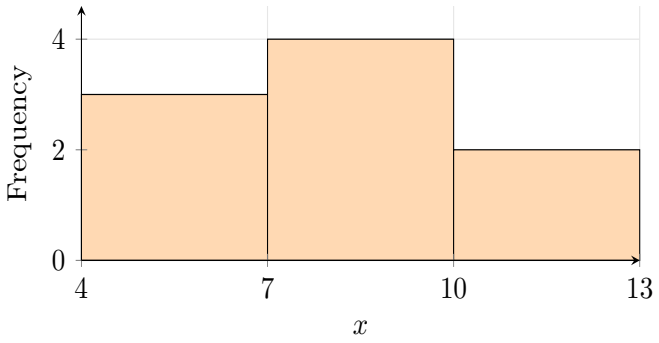
An equipment rental company monitors how long customers use rented equipment to evaluate the accuracy of its rental duration model. The company records the rental times of 100 transactions, measured in minutes. The results are summarized in the following frequency table, where each value represents an observed rental duration and its corresponding frequency.

x_i (min)	4	8	12	16	20	24
f_i	8	16	28	22	14	12

According to the company's theoretical model, the probability that a rental lasts at least 8 minutes but less than 20 minutes is 0.80. The company wants to assess how closely the observed rental durations agree with this prediction. Calculate the empirical probability of this interval and determine its absolute and percentage errors relative to the theoretical probability. **[1]**

8. Luggage masses, histogram data [C7-2]

An airline monitors the weight of passengers' carry-on luggage to evaluate the accuracy of its baggage weight model. During a routine inspection, airport staff record the masses of 9 randomly selected pieces of luggage, measured in kilograms. The individual measurements are no longer available, but their frequencies are preserved in the following histogram. Each bar represents the number of pieces of luggage whose masses fall within the corresponding interval, including the lower bound but excluding the upper bound.



According to the airline's theoretical model, luggage masses have a mean of $\mu = 9$ kg and a variance of $\sigma^2 = 7$ kg^2. The airline wants to assess how closely the observed luggage masses agree with these predictions. Using the grouped data, estimate the sample mean, sample variance, and sample standard deviation, then calculate the absolute and percentage errors of each statistic relative to its corresponding theoretical value. **[2]**

9. Chamber temperatures, raw data [C7-1]

A laboratory monitors the temperature inside a climate-controlled chamber to evaluate the accuracy of its temperature regulation model. During a routine inspection, technicians record five temperature readings, in degrees Celsius: $14, 17, 18, 19, 22$. According to the laboratory's theoretical model, the chamber temperature has a mean of $\mu = 19\,^\circ\text{C}$ and a variance of $\sigma^2 = 6\,^\circ\text{C}^2$. The technicians want to assess how closely the observed temperatures agree with these predictions. Calculate the sample mean, sample variance, and sample standard deviation, then determine the absolute and percentage errors of each statistic relative to its corresponding theoretical value. **[1]**

10. Parcels dispatched, frequency data [C7-1]

A delivery company monitors the number of parcels dispatched during short collection periods to evaluate the accuracy of its logistics forecasting model. During a routine assessment, the company records the number of parcels dispatched across nine collection periods. The results are summarized in the following frequency table, where each value represents the number of parcels dispatched and its corresponding frequency.

x_i	3	7	11
f_i	2	4	3

According to the company's theoretical model, the number of parcels dispatched per collection period has a mean of $\mu = 7$ parcels and a variance of $\sigma^2 = 9$ parcels^2. The company wants to assess how closely the observed dispatch quantities agree with these predictions. Calculate the sample mean, sample variance, and sample standard deviation, then determine the absolute and percentage errors of each statistic relative to its corresponding theoretical value. **[1]**

11. Walking times, grouped data [C7-2]

A university monitors how long students take to walk from the main entrance to their classrooms to evaluate the accuracy of its campus mobility model. During a routine assessment, staff record the walking times of nine students, measured in minutes. The results are summarized in the following frequency table, where each interval includes its lower bound but excludes its upper bound.

Time (min)	$[12, 18)$	$[18, 24)$	$[24, 30)$
f_i	2	3	4

According to the university's theoretical model, walking times have a mean of $\mu = 23$ minutes and a variance of $\sigma^2 = 30$ minutes^2. The university wants to assess how closely the observed walking times agree with these predictions. Using the grouped data, estimate the sample mean, sample variance, and sample standard deviation, then calculate the absolute and percentage errors of each statistic relative to its corresponding theoretical value. **[2]**